



Rick den Otter

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Education

Experimental Psychology, Utrecht University, PhD in Artificial Intelligence & Cognitive Neuroscience – Utrecht, the Netherlands May 2024 – present

My PhD is a continuation of my masters' thesis project, where I apply Hidden Multivariate Pattern analysis combined with deep spatiotemporal sequence models to decode cognitive processes from EEG at the trial level.

Utrecht University, MSc in Artificial Intelligence – Utrecht, the Netherlands Feb 2022 – Feb 2024

I conducted my thesis at the department of Experimental Psychology (Utrecht University) on classifying cognitive processes from EEG using machine learning classifiers. During this thesis I was chosen to present my research at the NVP Dutch Society for Brain and Cognition Winter Conference. My thesis was awarded a grade of 8.9, with a 9.5 for both the result and the process.

Hogeschool van Amsterdam, BSc in HBO-ICT Software Engineering – Amsterdam, the Netherlands Sept 2017 – June 2021

- Graduated Cum Laude (Grade: 8.3/10)
- Followed thematic semester "Big Data" and minor "Positive Psychology & Technology"
- Best first year project out of 150+ groups as voted by peers and teachers (2018)

Publications

Trial-level sequence modeling reveals hidden dynamics of dual-task interference May 2026

Rick den Otter, Anna Dame, Sjoerd Stuit, Leendert van Maanen
[10.1371/journal.pcbi.1014302](https://doi.org/10.1371/journal.pcbi.1014302) (PLOS Computational Biology)

Identifying trial-by-trial cognitive strategies in speed-accuracy trade-off through deep sequence modeling under review

Rick den Otter, Gabriel Weindel, Sjoerd Stuit, Leendert van Maanen

A Trial-Level Neural Predictor of Proactive Slowing and Stop-Signal Success in preparation

Rick den Otter, Leendert van Maanen, Leon Kenemans, Daan van Rooij

Grants and Awards

NWO Small Compute Grant (310.000 compute units on Snellius) Feb 2026

Rick den Otter

Conferences

CuttingGardens Groningen Sept 2026

Groningen, the Netherlands (Invited talk)

NVP Dutch Society for Brain and Cognition Winter Conference Dec 2025

Egmond aan Zee, the Netherlands (Poster presentation)

International Conference on Cognitive Neuroscience (ICON) Porto, Portugal (Poster presentation)	Sept 2025
Cognitive Computational Neuroscience (CCN) Amsterdam, the Netherlands (Poster presentation)	July 2025
Helmholtz Institute - Perception day Utrecht, the Netherlands (Poster presentation)	Nov 2024
NVP Dutch Society for Brain and Cognition Winter Conference Egmond aan Zee, the Netherlands (Poster presentation)	Dec 2023

Academic Service

(Co)-reviewer: eLife, Cognitive Computational Neuroscience conference

Teaching Experience

Machine Learning for Human Vision & Language (MSc Artificial Intelligence, Utrecht University):
Teaching work groups, grading exams

Human Network Analysis (MSc Applied Data Science, Utrecht University): Teaching work groups, grading exams

Programming (BSc Psychology, Utrecht University): Guiding and grading students during their programming assignments

Thesis supervision (BSc Artificial Intelligence, Utrecht University): Guiding and grading students during their BSc thesis

Thesis supervision (MSc Artificial Intelligence, Utrecht University): Guiding and grading students during their MSc thesis

Professional Experience

Software Engineer, Thesis Intern, Intern, Sweco Nederland – De Bilt, the Netherlands Jan 2019 – Feb 2024

My internships and job at Sweco Nederland started with developing and maintaining the REST API for Obsurv. Later, I developed a novel application handling concurrent long-running export tasks from a relational database to RDF triple graph format. Finally, I conducted my BSc thesis at Sweco Nederland on using machine learning to recognize defects in sewer inspection media. I integrated this model into a deployed product using Azure Machine Learning. Additionally, I guided multiple interns, assisting them with the technical part of their internships, and grading their theses. Technologies used: PL/SQL, Oracle stack, TypeScript, JavaScript, Express, NodeJS, Python, VR in ThreeJS, Azure for DevOps activities, Azure Machine Learning.

Courses and Summer Schools

Smoothening your writing process Utrecht, the Netherlands	July 2025
Inclusion and Social Safety training Utrecht, the Netherlands	Dec 2024
Summer School Advanced Communication Skills Utrecht, the Netherlands	Aug 2024

Skills

Programming: Proficient: Python (PyTorch, Pandas, NumPy), R (lme4), MATLAB (EEGLAB). Familiar: C#, JavaScript, HTML/CSS

Other technical skills: Git/GitHub, Linux, Docker, Distributed computing (Azure, Snellius), Adobe Creative Cloud (Photoshop, Illustrator, InDesign)

Experimental methods: EEG (BioSemi ActiveTwo)

Languages: English (fluent), Dutch (native)